

Projects

Custom Nix-Based Linux Distribution

I have **developed** and **daily driven** a custom **Nix-based Linux distribution**, tailored for **personal productivity** and system requirements. This involved configuring a **reproducible** and **declarative** environment using Nix, optimising **package management**, and fine-tuning system services for **stability** and **performance**. I automated system provisioning with custom **Nix derivations**, ensuring **seamless updates**, **rollback capabilities** and atomicity of **program configurations**. Through this project, I gained deep expertise in **functional package management**, system **configuration**, and **Linux internals**, enhancing my ability to build and maintain **reliable, scalable** computing environments.

Web-App written in pure Rust

As part of a three-person team, I developed a **high-performance web application** for an online store using pure **Rust** with the **Leptos** framework. My contributions included building **reactive UI** components, integrating a **robust backend** with **Actix** for efficient data handling, and management of the **build and deployment system** through **Nix**. By leveraging Rust's **memory safety** and **concurrency features**, we ensured a highly responsive and secure shopping experience. The project emphasized **clean architecture**, **minimal runtime overhead**, and a fully **type-safe** codebase, resulting in a **fast and scalable** e-commerce platform. This experience deepened my expertise in **full-stack Rust development** and **modern web technologies**.

Work Experience

MERN Full Stack Web Developer

Bar Monkey Calisthenics: Jan 24 > July 24

For **six months**, I worked as a **solo full-stack web developer** at **Bar Monkey**, where I designed, developed, and deployed a web application using the **MERN (MongoDB, Express, React, Node.js)** stack. I built a **dynamic and responsive** frontend with React while implementing a robust backend with **Express and MongoDB**, ensuring seamless data flow and API performance. Deployment was handled via **Docker**, containerising the application for **scalability** and **reliability** on fly.io. Throughout the project, I managed the **full development life-cycle**, from **planning** and **coding** to **deployment** and **maintenance**, gaining valuable experience in **end-to-end web development** and **cloud-based infrastructure**.

Data Management Intern

AIB: June 23 > August 23

During my **summer internship** at AIB, I worked on modernising **data management processes** for the lending division. My primary responsibility was converting a **large volume** of **historical paper records** into structured digital formats, such as **Excel spreadsheets** and **database-ready files**. This involved meticulous **data entry, validation**, and organisation to ensure accuracy and consistency. Additionally, I helped **streamline data workflows**, making retrieval and analysis more efficient for the team. Through this experience, I developed strong skills in **data handling, digitization strategies**, and **process optimization** within a financial services context.

Education

Institution	Goal	Dates Attended	Outcome
Trinity College Dublin	B.A. Computer Science	Sep 23 > May 27	81% Grade Average
Clonkeen College	Leaving Cert	Sept 18 > June 23	601 Points

Skills

Rust, DevOps, Nix, SQL, Web, React, JavaScript, TypeScript, MongoDB, Express, Node.js, Docker, Clean Code, Type Safety